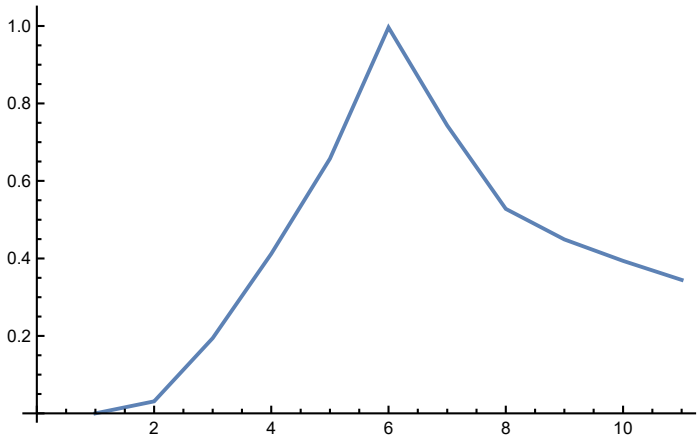
In[ ]:=
```
(* Random phase JONSWAP spectrum with modal frequency omegam*)
```

In[ ]:=
```
Table[Om[i], {i, -5, 5}]
```

Out[ ]=
```
{0.5, 0.6, 0.7, 0.8, 0.9, 1., 1.1, 1.2, 1.3, 1.4, 1.5}
```

In[ ]:=
```
ListLinePlot[Table[A[i], {i, -5, 5}]]
```

Out[ ]=

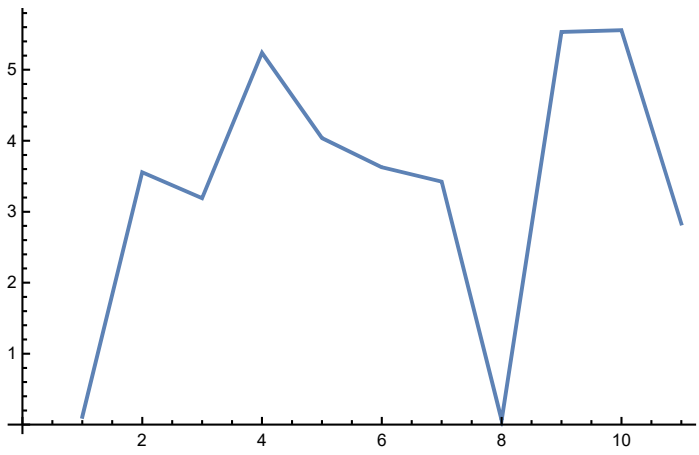


In[ ]:=
```
Table[A[i], {i, -5, 5}]
```

Out[ ]=
```
{0.000275988, 0.0310087, 0.194104, 0.411984, 0.657557,
 0.996291, 0.743132, 0.527962, 0.449042, 0.393818, 0.344679}
```

In[ ]:=
```
ListLinePlot[Table[CC[i], {i, -5, 5}]]
```

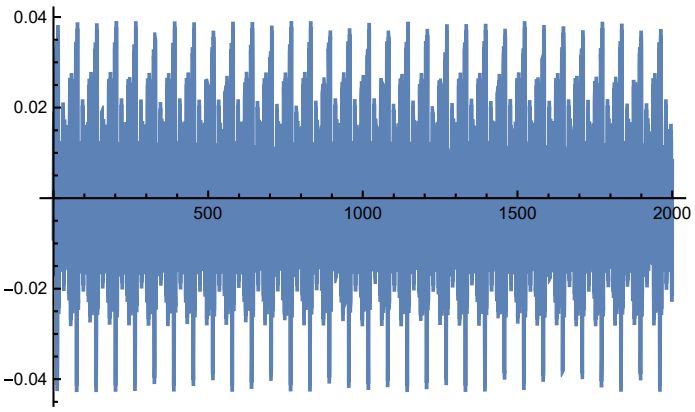
Out[ ]=



In[ ]:=
```
gg[t_] := Sum[A[i] × Cos[Om[i] t + CC[i]], {i, -5, 5}]
```

In[•]:=
```
Plot[-eps gg''[t], {t, 0, 2000}]
```

Out[•]=



In[•]:=
```
eqz1ent := z1''[t] + z1[t] + eps lam (z1'[t] - z2'[t]) + eps k (z1[t] - z2[t]) == -eps gg''[t]
```

In[•]:=
```
eqz2ent := z2''[t] + z2[t] + eps lam (z2'[t] - z1'[t]) + eps k (z2[t] - z1[t]) +
   eps lama (z2'[t] - vz'[t]) + eps ka (z2[t] - vz[t])^3 == - eps gg''[t]
```

In[•]:=
```
eqz3ent :=
 eps vz''[t] - eps lama (z2'[t] - vz'[t]) - eps ka (z2[t] - vz[t])^3 == - eps^2 gg''[t]
```

In[•]:=
```
solnzent = NDSolve[ {eqz1ent, eqz2ent, eqz3ent, z1[0] == 0, z1'[0] == 0,
   z2[0] == 0, z2'[0] == 0, vz[0] == 0, vz'[0] == 0}, {z1, z2, vz}, {t, 0, 2000}]
```

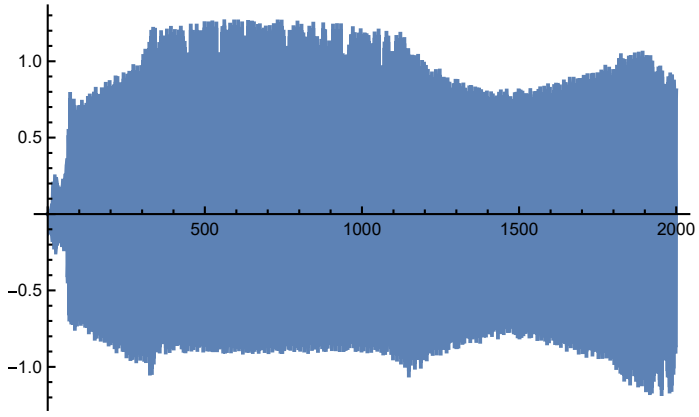
Out[•]=

```
{{z1 → InterpolatingFunction[ ⊞ Domain: {{0., 2.00 × 10^3}}
                              Output: scalar ],
  z2 → InterpolatingFunction[ ⊞ Domain: {{0., 2.00 × 10^3}}
                              Output: scalar ],
  vz → InterpolatingFunction[ ⊞ Domain: {{0., 2.00 × 10^3}}
                              Output: scalar ]}}
```

In[•]:=
```
(*k=1; ka=14/3; lam=0.2; lama=0.2, eps=0.01 *)
```

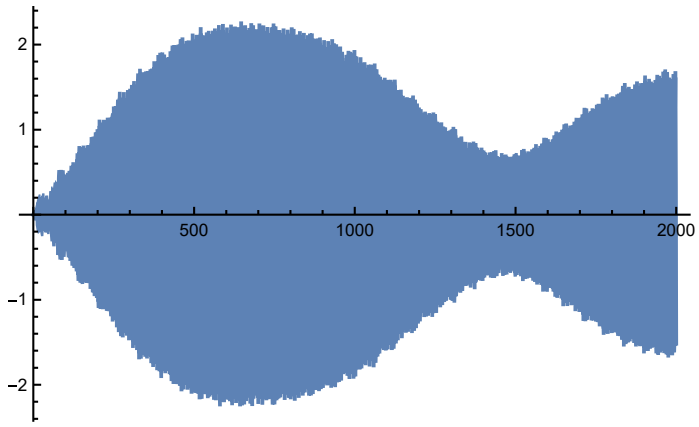
In[ ]:=
```
plotwz2 = Plot[Evaluate[ {z2[t] - vz[t]} /. solnzent], {t, 0, 2000}, PlotRange → All]
```

Out[ ]=



In[ ]:=
```
Plotuz = Plot[Evaluate[ {z2[t] + eps vz[t]} /. solnzent], {t, 0, 2000}, PlotRange → All]
```

Out[ ]=



In[ ]:=
```
wz1[t_] := Evaluate[z1[t] - z2[t] /. solnzent]
```

In[ ]:=
```
wz2[t_] := Evaluate[z2[t] - vz[t] /. solnzent]
```

In[ ]:=
```
uz[t_] := Evaluate[z2[t] + eps vz[t] /. solnzent]
```

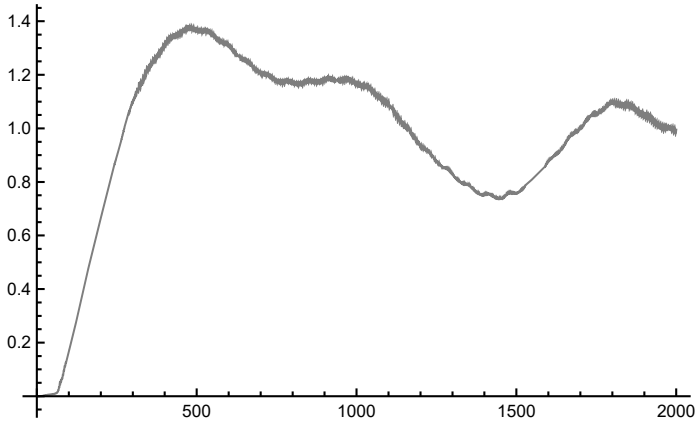
In[ ]:=
```
Phiz1[t_] := wz1'[t] + I wz1[t]
```

In[ ]:=
```
Phiz2[t_] := wz2'[t] + I wz2[t]
```

In[ ]:=
```
Phiz3[t_] := uz'[t] + I uz[t]
```

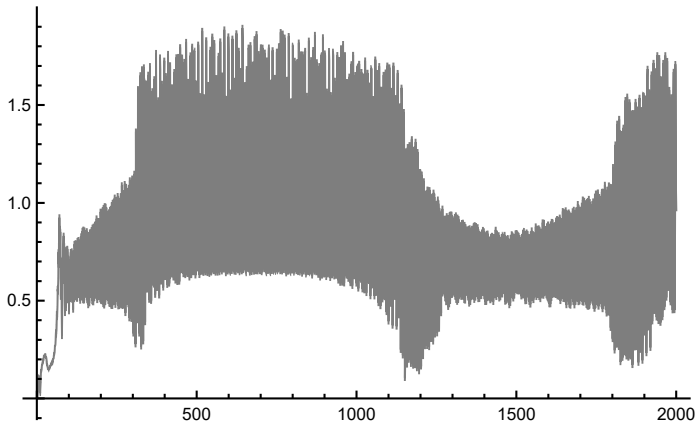
In[ ]:=
```
pExactPhiz1 =
 Plot[Abs[Phiz1[t]], {t, 0, 2000}, PlotRange → All, PlotStyle → {Thickness[0.001], Gray}]
```

Out[ ]=



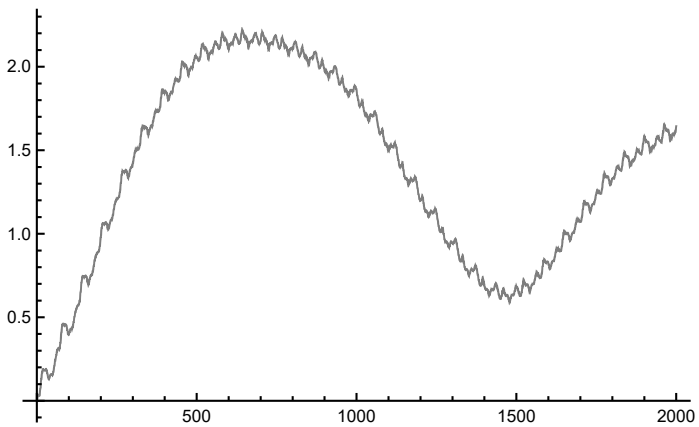
In[ ]:=
```
pExactPhiz2 =
 Plot[Abs[Phiz2[t]], {t, 0, 2000}, PlotRange → All, PlotStyle → {Thickness[0.001], Gray}]
```

Out[ ]=



In[ ]:=
```
pExactPhiz3 =
 Plot[Abs[Phiz3[t]], {t, 0, 2000}, PlotRange → All, PlotStyle → {Thickness[0.001], Gray}]
```

Out[ ]=

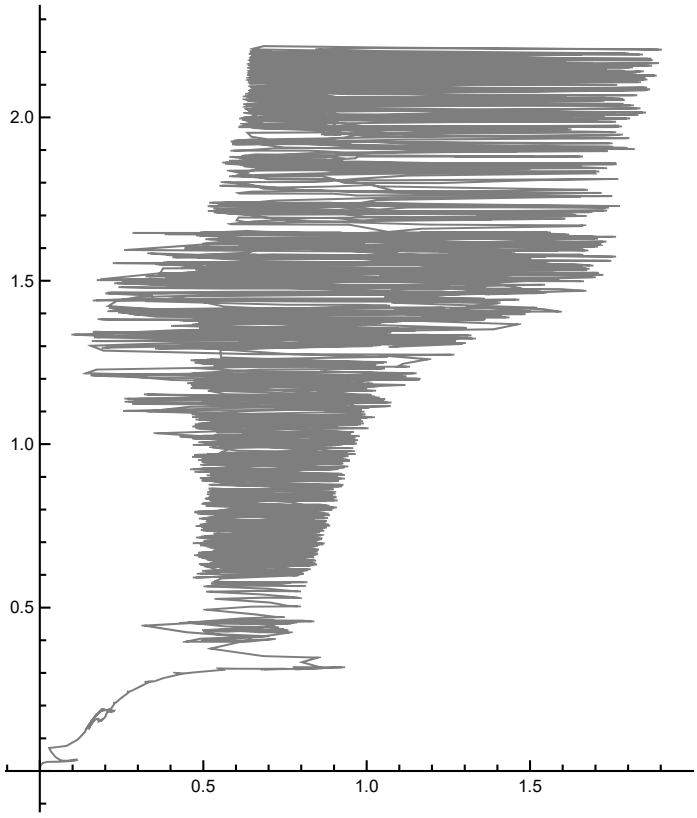


In[ ]:=
```
ExactPhizProj = ParametricPlot[Evaluate[ {intPz2[t], intPz3[t]} /. solnzent],
  {t, 0, 2000}, PlotStyle → {Thickness[0.001], Gray}, PlotRange → All]
```

InterpolatingFunction: Input value {0.0408163} lies outside the range of data in the interpolating function. Extrapolation will be used.

InterpolatingFunction: Input value {0.0408163} lies outside the range of data in the interpolating function. Extrapolation will be used.

Out[ ]=

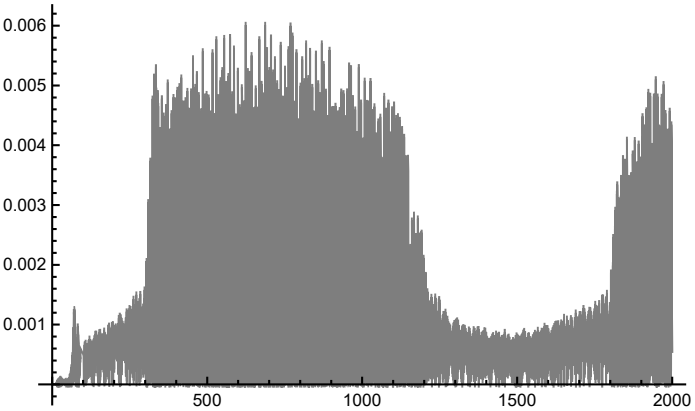


In[ ]:=
```
(***************TTotal power dissipated in the system*********************)
```

In[ ]:=
```
(* Nonlinear system *)
```

In[ ]:=
```
pztotnonl = Plot[Evaluate[ eps lama (z2'[t] - vz'[t])^2 /. solnzent],
  {t, 0, 2000}, PlotRange → All, PlotStyle → {Thickness[0.001], Gray}]
```

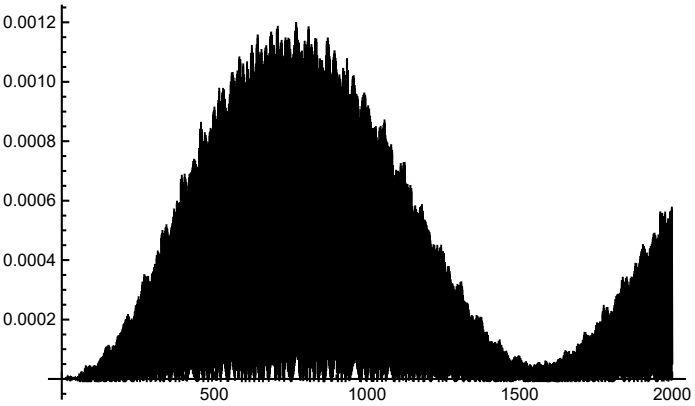
Out[ ]=



In[ ]:=
```
(* Linear system *)
```

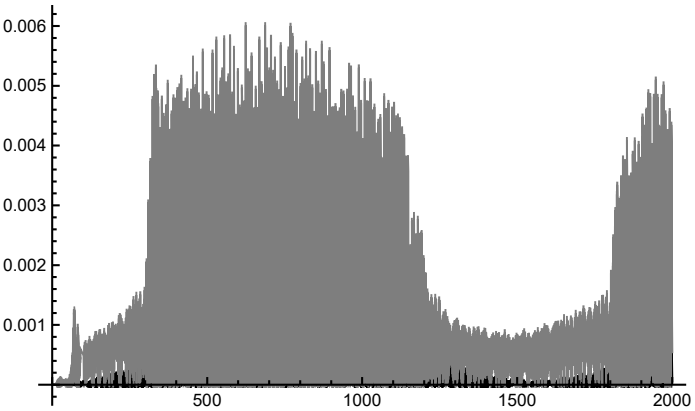
In[ ]:=
```
pztotlin = Plot[Evaluate[ eps lama (z2'[t] - vz'[t])^2 /. solnzent],
  {t, 0, 2000}, PlotRange → All, PlotStyle → {Thickness[0.001], Black}]
```

Out[ ]=



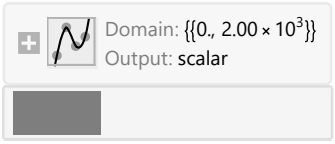
In[ ]:=
```
Show[pztotlin, pztotnonl]
```

Out[ ]=



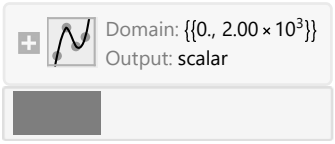
```
In[•]:= (*****************Slow Flow Analysis and
         comparison to Exact Solution*******************)

In[•]:= eqP1 := P1'[t] - eps k I P1[t] - eps (lama / 2) P2[t] +
           (3 / 8) eps ka I Abs[P2[t]]^2 × P2[t] + (eps lam / 2) P1[t] == 0

In[•]:= eqP2 :=
         P2'[t] + (I / (2 (1 + eps))) (P2[t] - P3[t]) + eps (k I / 2) P1[t] + (((1 + eps) lama ) / 2) P2[t] -
           (3 / 8) (1 + eps) ka I Abs[P2[t]]^2 × P2[t] + (eps lam / 2) P3[t] == 0

In[•]:= eqP3 :=
         P3'[t] + (I eps / (2 (1 + eps))) (P3[t] - P2[t]) + I eps (k / 2) P1[t] + (eps lam / 2) P3[t] ==
           (1 + eps) eps 0.9962909987148387 (1 / 2) Exp[I 3.626268087926144]

In[•]:= solnP =
         NDSolve[ {eqP1, eqP2, eqP3, P1[0] == 0, P2[0] == 0, P3[0] == 0}, {P1, P2, P3}, {t, 0, 2000}]

Out[•]=
```

$$\left\{\left\{P1 \to \text{InterpolatingFunction}\left[\right.\right.\right.$$



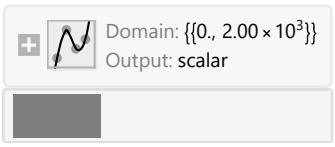
$$\left.\left.\left.\right],\right.\right.$$

$$P2 \to \text{InterpolatingFunction}\left[\right.$$



$$\left.\right],$$

$$P3 \to \text{InterpolatingFunction}\left[\right.$$



$$\left.\right]\right\}\right\}$$

In[ ]:= **Show[PhiProj, ExactPhizProj]**

Out[ ]=